

Yonghyeon Gwon

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EDUCATION

M.S. in Neuroscience (Expected)

Sep 2024–Aug 2026

Seoul National University, Interdisciplinary Program in Neuroscience

Seoul, Korea

· Overall GPA: 4.07/4.3

B.S. in Biological Science and B.A in Linguistics (double major)

Feb 2018– Aug 2024

Seoul National University, Department of Biological Science and Linguistics

Seoul, Korea

· Overall GPA: 3.81/4.3

RESEARCH INTERESTS

Neurolinguistics via invasive recording and OPM-MEG; Lexical to sentence integration; Overt and imagined speech decoding; iEEG foundation model

RESEARCH EXPERIENCE

Human Brain Function Laboratory

Jan 2024–Present

Seoul National University

· Advisor: Prof. Chun Kee Chung

Graduate Student

Aug 2024–Aug 2026

Undergraduate Research Assistant

Sep 2022–Aug 2023

Neurolinguistics and Speech Decoding Research

- Designed language-related experiments using invasive recordings and conducted data collection.
- Investigated whether word-level and sentence-specific components of iEEG sentence representations can be dissociated spatiotemporally immediately after question presentation.
- Developed a computational framework extending Rational Speech Act models through the Free Energy Principle and active inference (co-advised by Dr. C. Elliot Murphy at UThealth).
- Developed a flow-matching neural-to-speech synthesis model conditioned on phoneme-decoding outputs derived from iEEG recordings during speech production.
- Investigated whether the neural representation of denasalized Korean /n/ reflects an articulation-based /t/-like state or the abstract phonemic representation [n].
- Installed and configured an OPM-MEG system; designed a word-level overt and imagined-speech paradigm and recorded data.

NeuroAI Research

- Developed a Denoising Diffusion Probabilistic Model (DDPM) to generate iEEG signals conditioned on somatosensory states (co-advised by Prof. Yung Kyun Noh, Hanyang University).
- Developed DIVER-1, a state-of-the-art iEEG foundation model; performed preprocessing of continuous 24/7 recordings, model-component development and pretraining, and downstream evaluation (co-advised by Prof. Jiook Cha, Seoul National University). Contributed to DIVER-CLIP, a contrastive-learning framework that jointly aligns iEEG with both speech and text embeddings.

Funded Research Projects

- Participated in clinical testing of a newly developed invasive electrode, including SNR analysis and on-site recording.
- Served as the primary working-level coordinator for the Brain-to-X theme of the Alchemist Project., (annual budget: 4 billion). Coordinated hardware teams working on electrodes, chips, batteries, and communications with software researchers, and led administrative work including report compilation.

Undergraduate Research Support Program

Seoul National University (Advisor: Prof. Hyeyoung Shin)
Research Assistant

Mar 2024– Jun 2024

- Analyzed spike-train autocorrelograms for individual units and cross-correlograms between units in the rodent visual cortex and LGN during paired illusory visual stimulation; identified periodic broadband-gamma activity.

Seoul National University (Advisor: Prof. Hyoung. F. Kim)
Research Assistant

Jul 2020–Dec 2020

- Assisted with RNA sequencing of the rodent basal ganglia and comparative analysis against existing macaque RNA-sequencing data.

PUBLICATIONS AND MANUSCRIPTS

Gwon, Y., & Chung, C. K. (2026). *Distinct post-sentence neural patterns representing lexical items vs. sentence integration*. NeuroImage, 121703. doi: [10.1016/j.neuroimage.2026.121703](https://doi.org/10.1016/j.neuroimage.2026.121703)

Kim, S., Han, J., Han, D. D., Lee, A. L., Lee, J., **Gwon, Y.**, Paris, Z., Chung, C. K., Bahk, S., Kim, N. S., Hwang, S. J., & Cha, J. (2026). *BAT-CLIP: Trimodal alignment of brain, audio, and text*. MLSP 2026. **Accepted**

Han, D. D[†], **Gwon, Y.**[†], Lee, A. L[†], Lee, T[†], Lee, S. J., Choi, J., Lee, S., Bang, J., Lee, S., Park, D. K., Yoo, S., Chung, C. K., & Cha, J. (2025). *DIVER-1: Scaling intracranial EEG foundation models for transferable representations* [Preprint]. arXiv. <https://arxiv.org/abs/2512.19097>

Gwon, Y., Murphy, E., & Chung, C. K. *A Coupled Active-Inference Extension of Rational Speech Act Models* [Manuscript in preparation].

Gwon, Y., Roh, S., Park, Y. J., Lee, E., Kang, H.-G., & Chung, C. K. *Uncertainty-Aware Neural-to-Speech Synthesis with Flow Matching* [Manuscript in preparation].

[†] Equal contribution.

PRESENTATIONS

Gwon, Y., Lee, E., & Chung, C. K. (2026). *Generative model-based neural-to-speech synthesis using confidence-weighted phoneme decoding outputs*. Poster accepted for presentation at Neuroscience 2026, the annual meeting of the Society for Neuroscience, Washington, DC.

Gwon, Y., & Chung, C. K. (2025). *Neural encoding of Korean /n/ allophones: Distinguishing articulatory and abstract phonemic representations*. Poster presented at Neuroscience 2025, the annual meeting of the Society for Neuroscience, San Diego, CA.

Gwon, Y., & Chung, C. K. (2024). *Decoding the “living or non-living” and “body parts or non-body parts” semantic categories from intracranial EEG data*. Poster presented at Neuroscience 2024, the annual meeting of the Society for Neuroscience, Chicago, IL

TEACHING EXPERIENCE

Volunteer Tutor

May 2018–Jul 2020

Provided weekly academic mentoring to socioeconomically disadvantaged elementary & middle school students.

SKILLS

Programming Tools: Python, MATLAB

Languages: Korean, native; English, professional working proficiency